Straight Line

Mind map

Geometry basics

- Parallelogram: diagonals bisect
- Rhombus $p_1 = p_2$; cyclic $A + C = \pi$

Coordinates and distance

- $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- Polar $(r\cos\theta, r\sin\theta)$

Section formula

- $\dfrac{mx_2 \pm nx_1}{m \pm n}$
- Harmonic: AP, AB, AQ in H.P.

Triangle centres

- $G,\ I,\ O,\ H$ formulas
- $G = \dfrac{H + 2O}{3}$
- Isosceles: collinear

Area

- $\frac{1}{2}|\det|$; $0 \Rightarrow$ collinear
- Polygon: shoelace

Locus

- $P(h, k)$, eliminate
- Circle / conics from distances

Forms of a line

- $y = mx + c$, $\dfrac{x}{a} + \dfrac{y}{b} = 1$
- $x\cos\alpha + y\sin\alpha = p$
- Parametric r

Angle, foot, image

- $\left|\dfrac{m_1 - m_2}{1 + m_1 m_2}\right|$
- Foot / image ratio form

Distances

- $\dfrac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$
- Parallel: $\dfrac{|c_1 - c_2|}{\sqrt{a^2 + b^2}}$

Position and concurrency

- Same side: $L(P)L(Q) > 0$
- Concurrent: $\det = 0$

Images

- $(a, b) \to (b, a)$ in $y = x$
- Line image via a point

Family of lines

- $L_1 + \lambda L_2 = 0$
- Linear parameter: fixed point

Axes changes

- $x = X + h$
- $x = X\cos\theta - Y\sin\theta$

Angle bisectors

- $\dfrac{L_1}{\sqrt{\ }} = \pm\dfrac{L_2}{\sqrt{\ }}$
- Acute: $a_1a_2 + b_1b_2 < 0$ test

Pair of lines

- $\tan\theta = \left|\dfrac{2\sqrt{h^2 - ab}}{a + b}\right|$
- $\Delta = 0$; $a + b = 0$ perpendicular

Homogenisation

- $1 = \dfrac{lx + my}{n}$
- Gives OA, OB pair